

Lecture 01: 10 Usability Heuristics

COMS 4170 • Spring 2026

Prof. Celeste Layne
Tuesday January 20, 2026



Usability Heuristics

The most-used usability heuristics for user interface design were developed by Jakob Nielsen (1994)



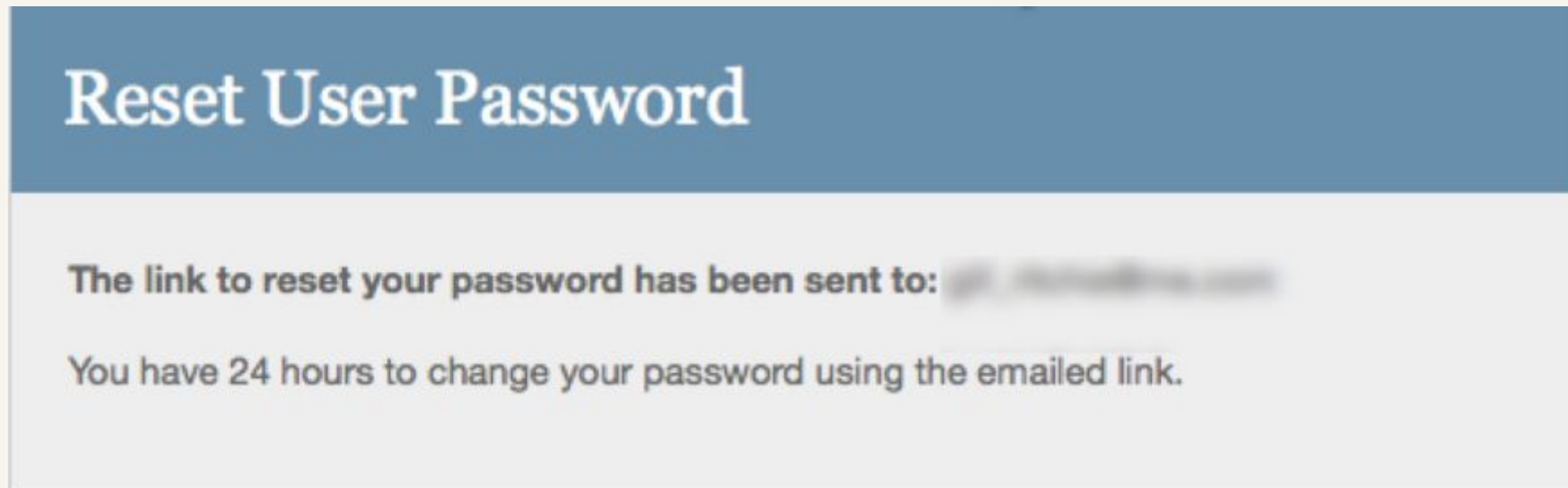
1. Visibility of System Status

The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.



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VIOLATION

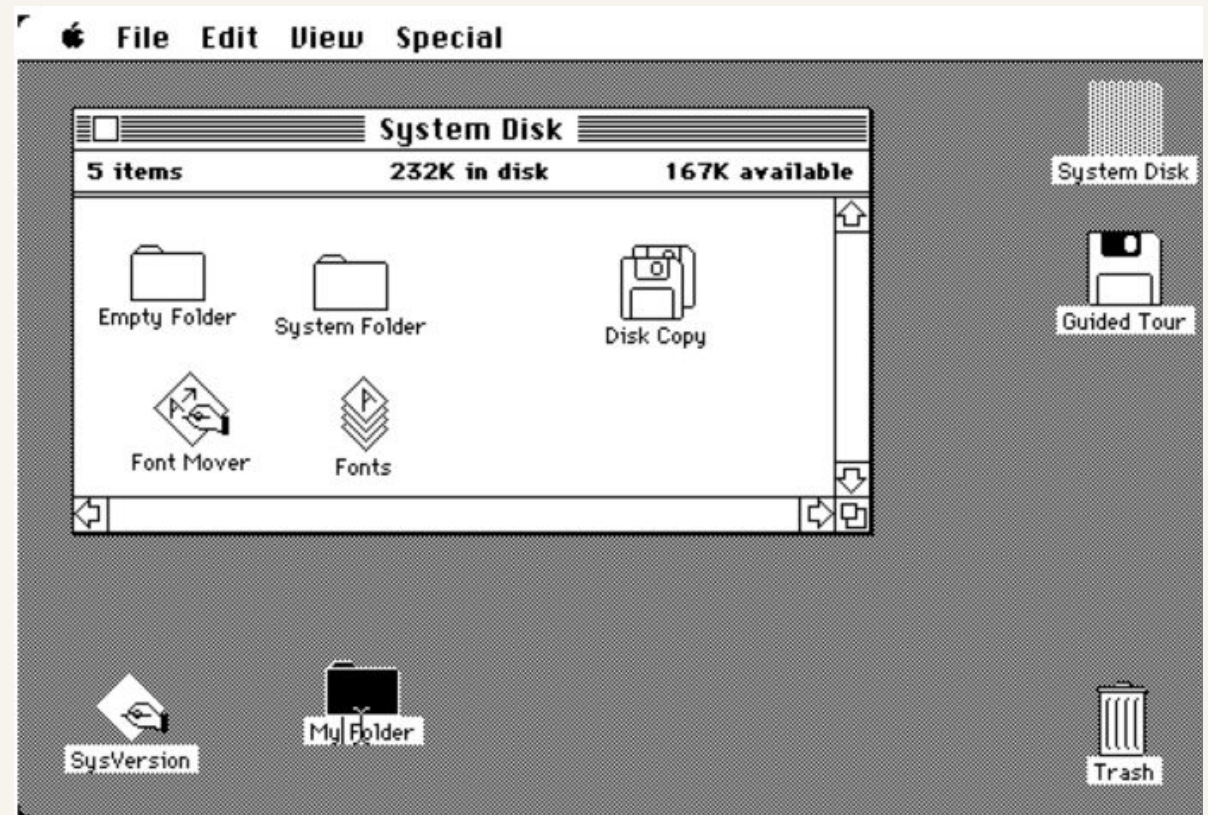


4 hours later



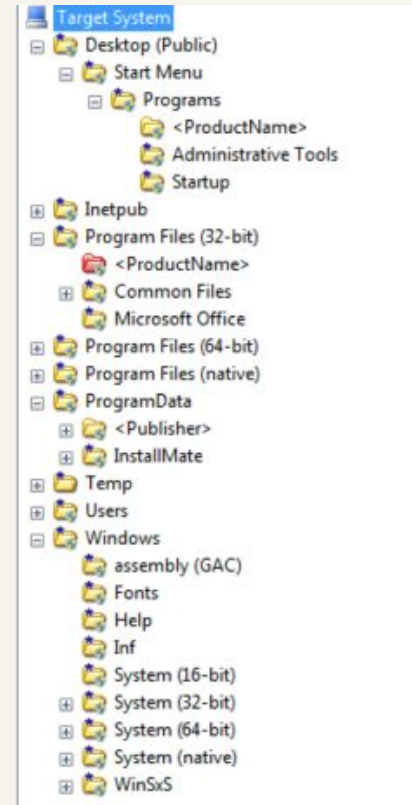
2. Match between system and the real world

The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms.

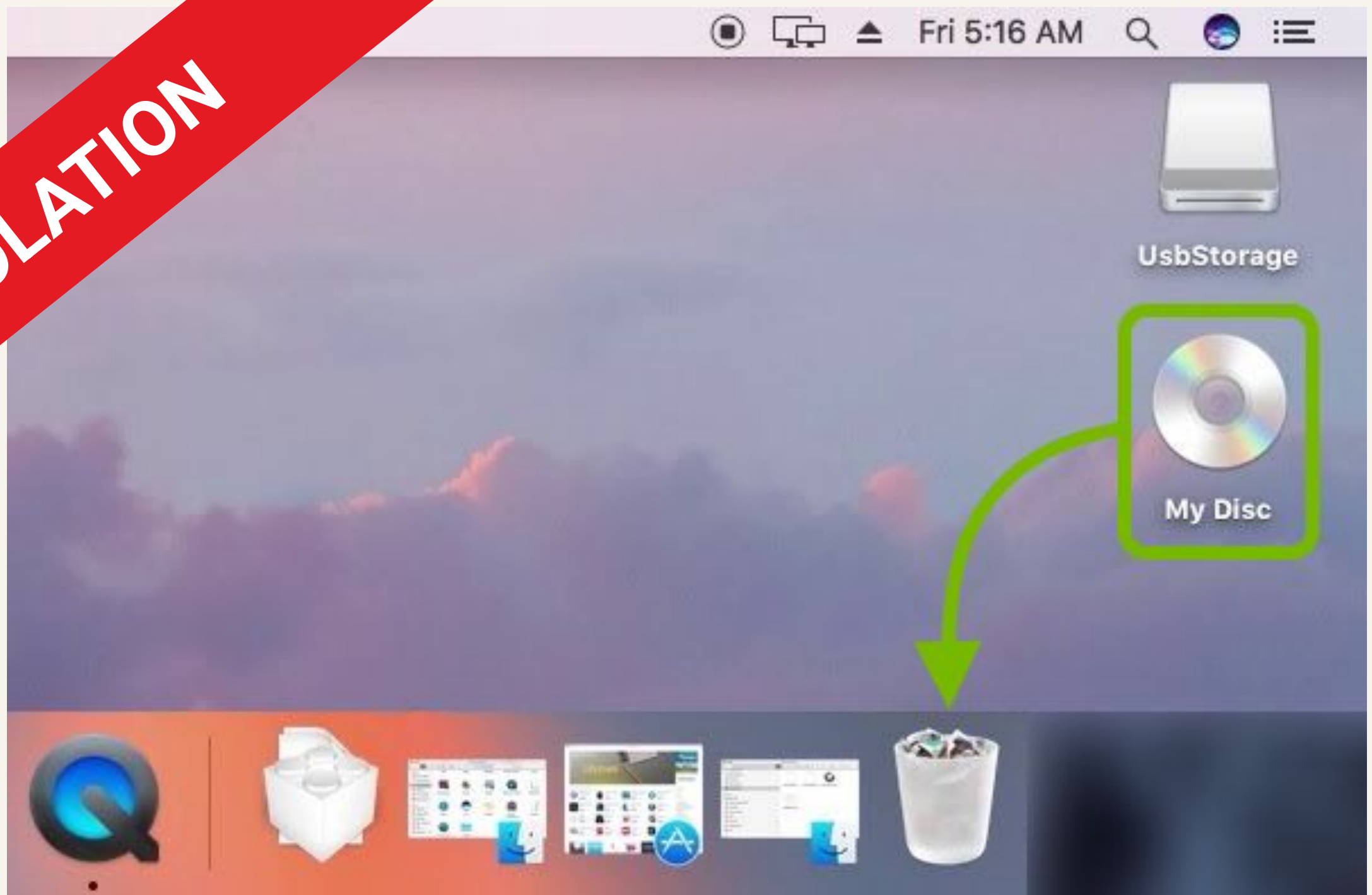


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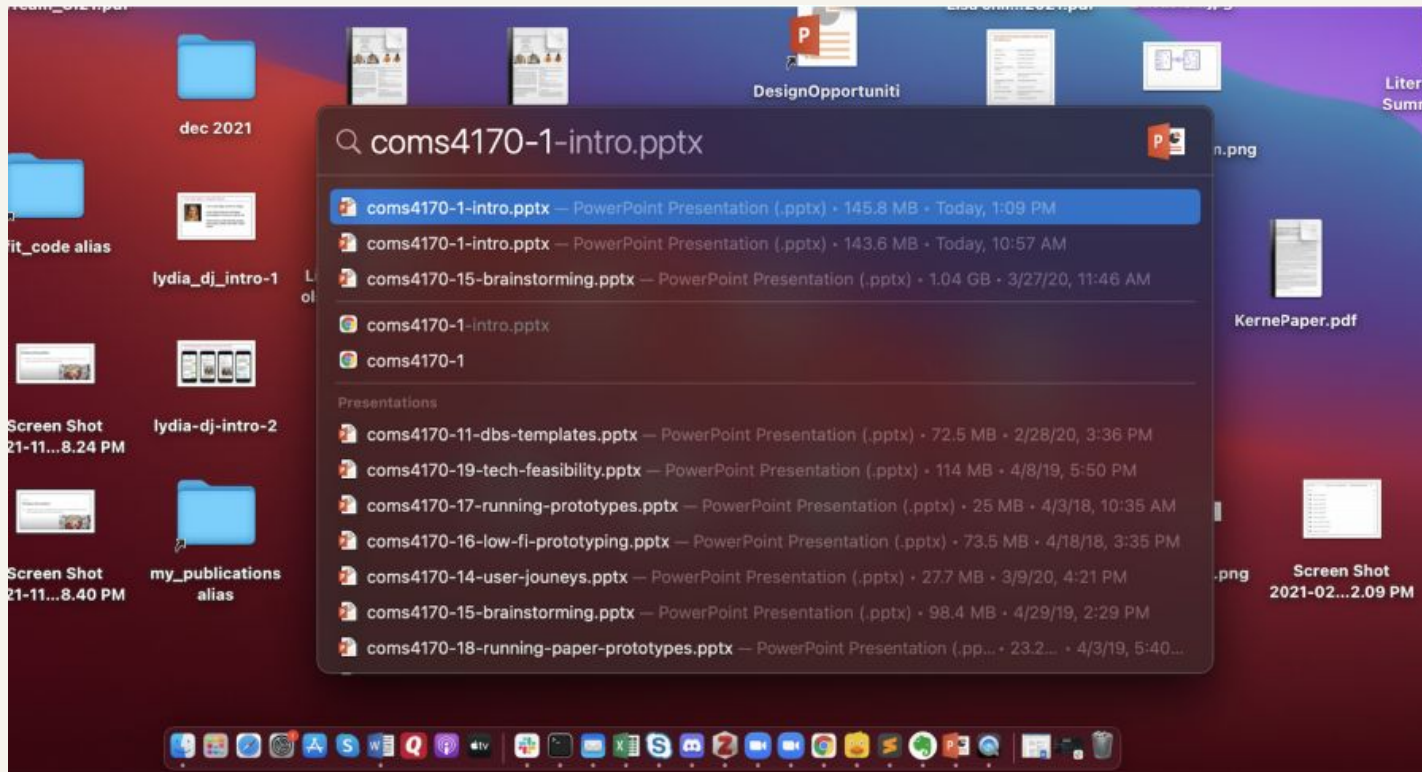


VIOLATION



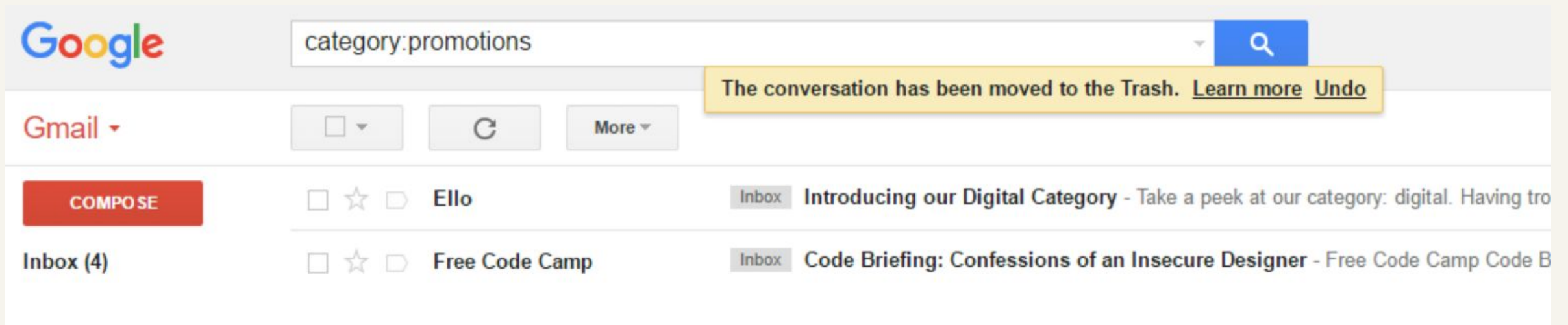
3. User control and freedom (Navigation)

Users should be able to quickly make choices, correct mistakes or backtrack on choices made. Support undo and redo.



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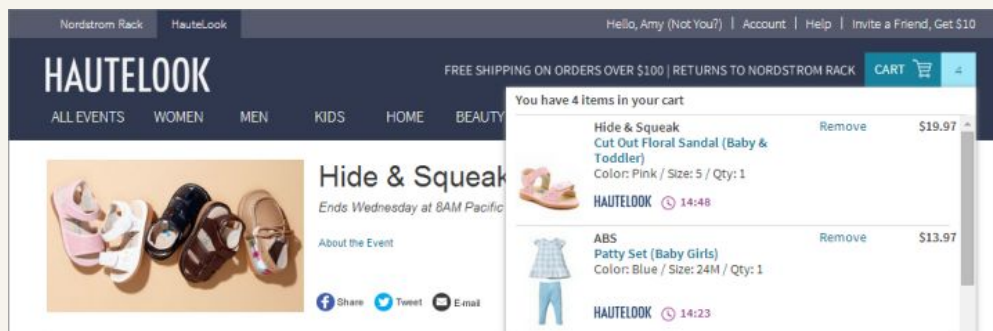
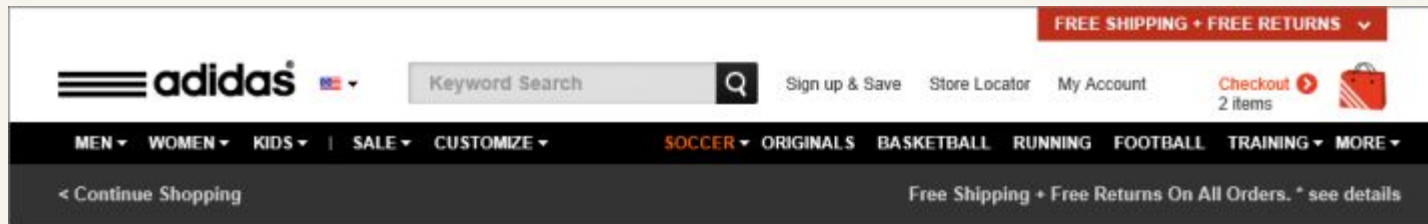
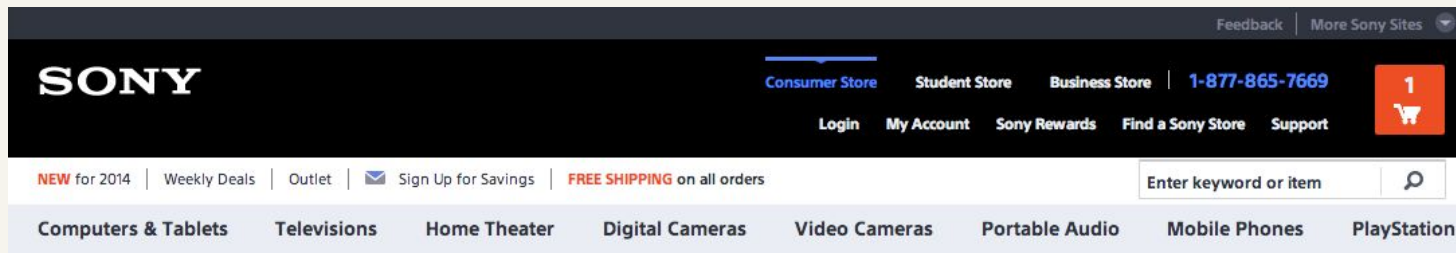


4. Consistency and standards

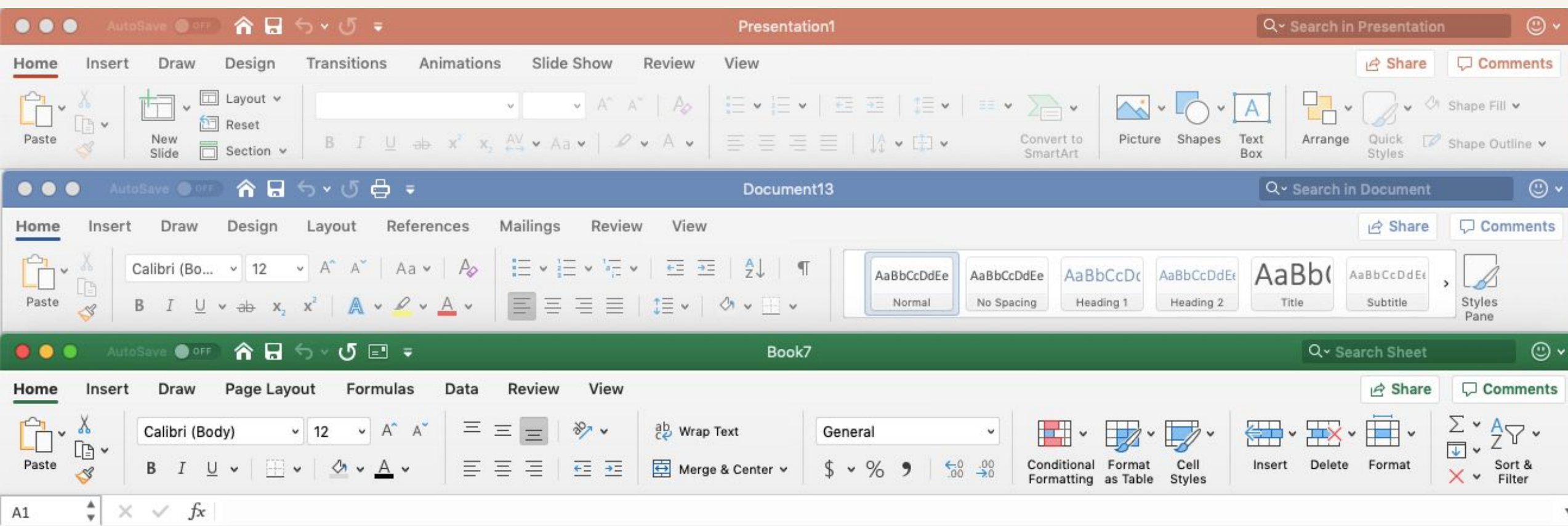
Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.



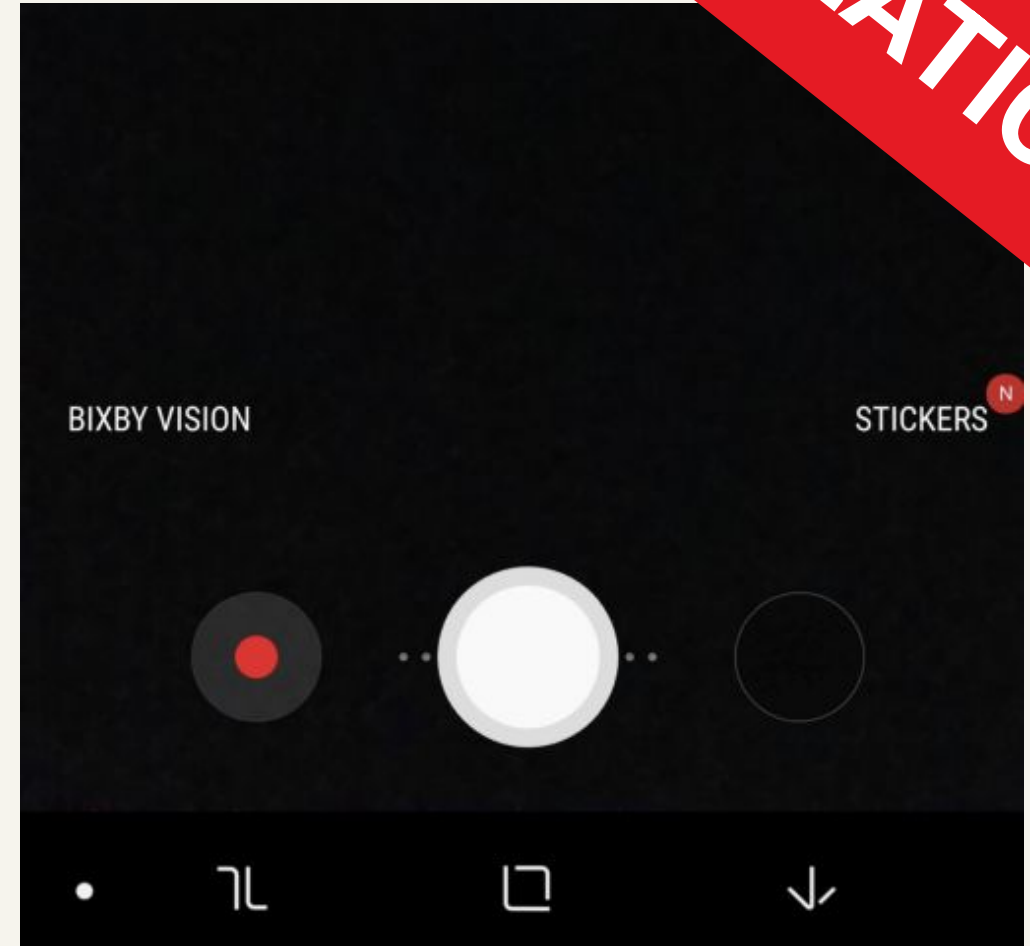
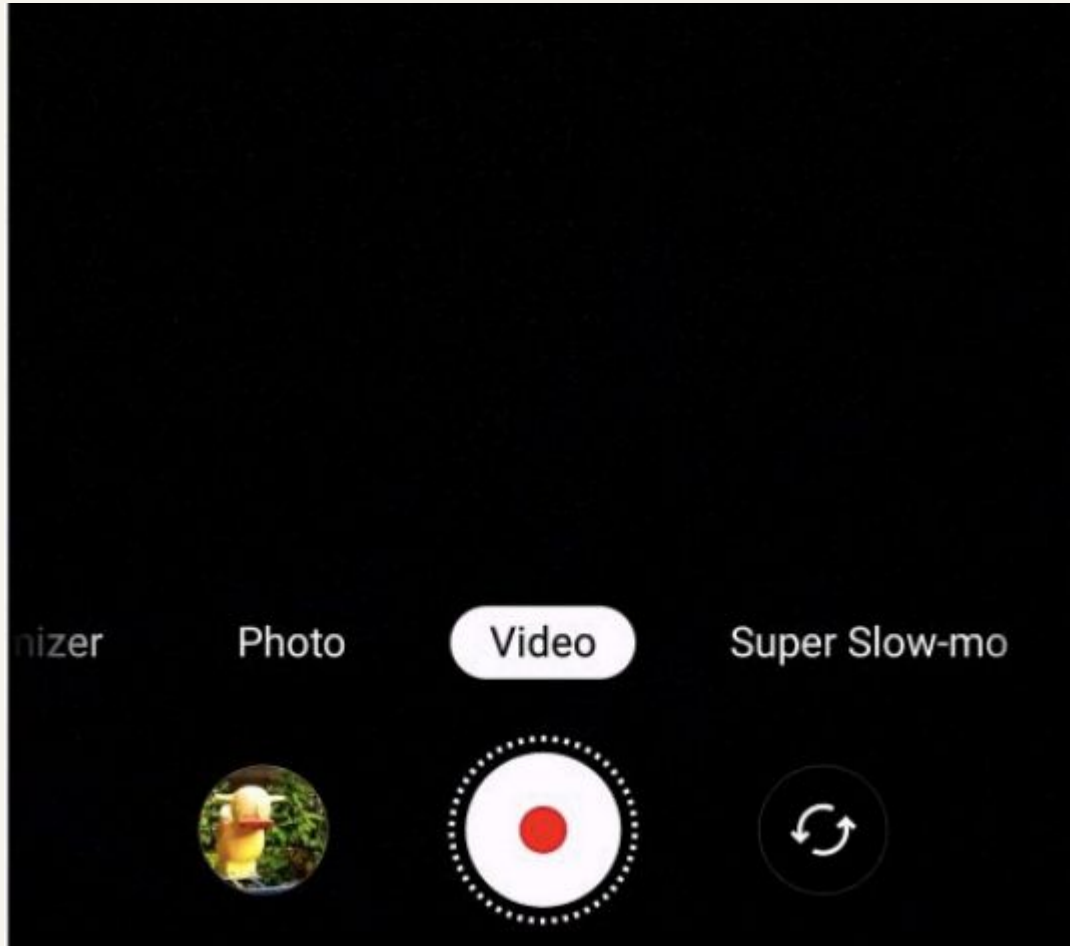
4. Consistency and standards



4. Consistency and standards



4. Consistency and standards



VIOLATION



5. Error prevention

Even better than good error messages is a careful design which prevents a problem from occurring in the first place.



mail.google.com says

It seems like you forgot to attach a file.

You wrote "I have attached" in your message, but there are no files attached. Send anyway?

Cancel

OK

Compose

Pause Inbox

Inbox 41

Starred

Snoozed



- ☐ ☆ Aman Gautam
- ☐ ☆ Amari Gautam
- ☐ ☆ Aman, me, Anish 3
- ☐ ☆ Aman Gautam 2
- ☐ ☆ Aman Gautam
- ☐ ☆ Gussie Zapoticky 2
- ☐ ☆ Anish Kumar 2
- ☐ ☆ Gaurav Jassal
- ☐ ☆ Anish Kumar

I have attached the images for your reference

Sans Serif



B

I

U

A



Send



Send Later



in 2 days

if no reply



Responsible



Gaurav Jassal





Select depart date

July 2015

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

August 2015

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

Book

Round trip

BOS

Boston Logan, MA - BOS

LGA

New York (LaGuardia), NY -
LGA

07/08

Depart Wed, Jul 8 2015

07/11

Return Sat, Jul 11 2015

1

Adults

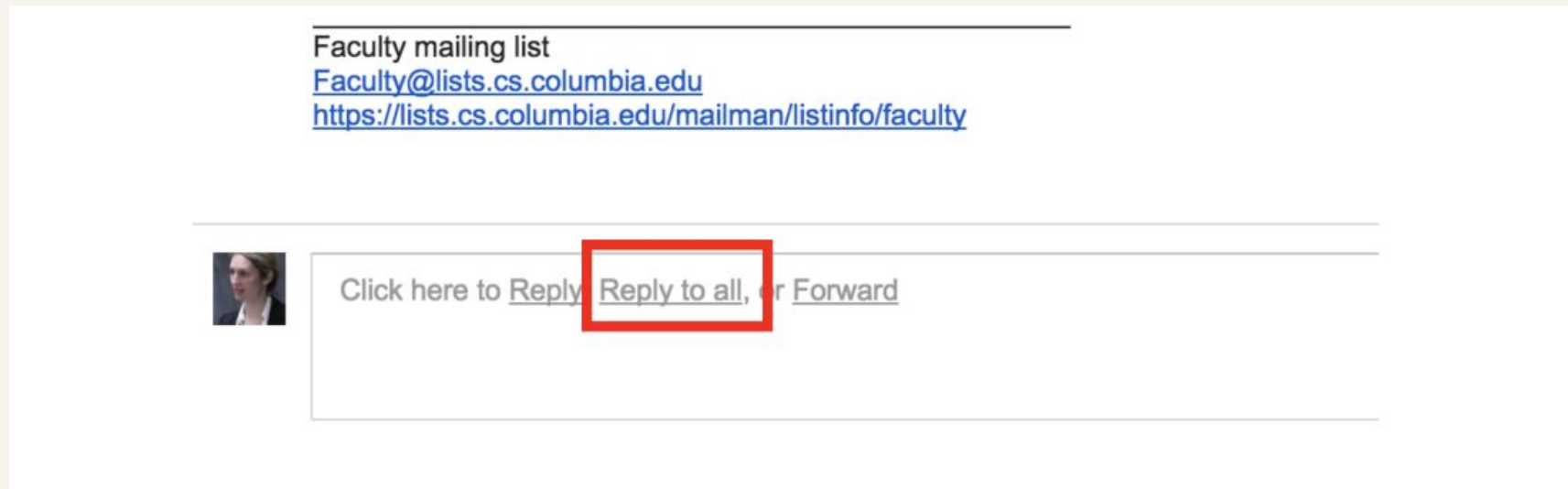
0

Seniors

5. Error prevention

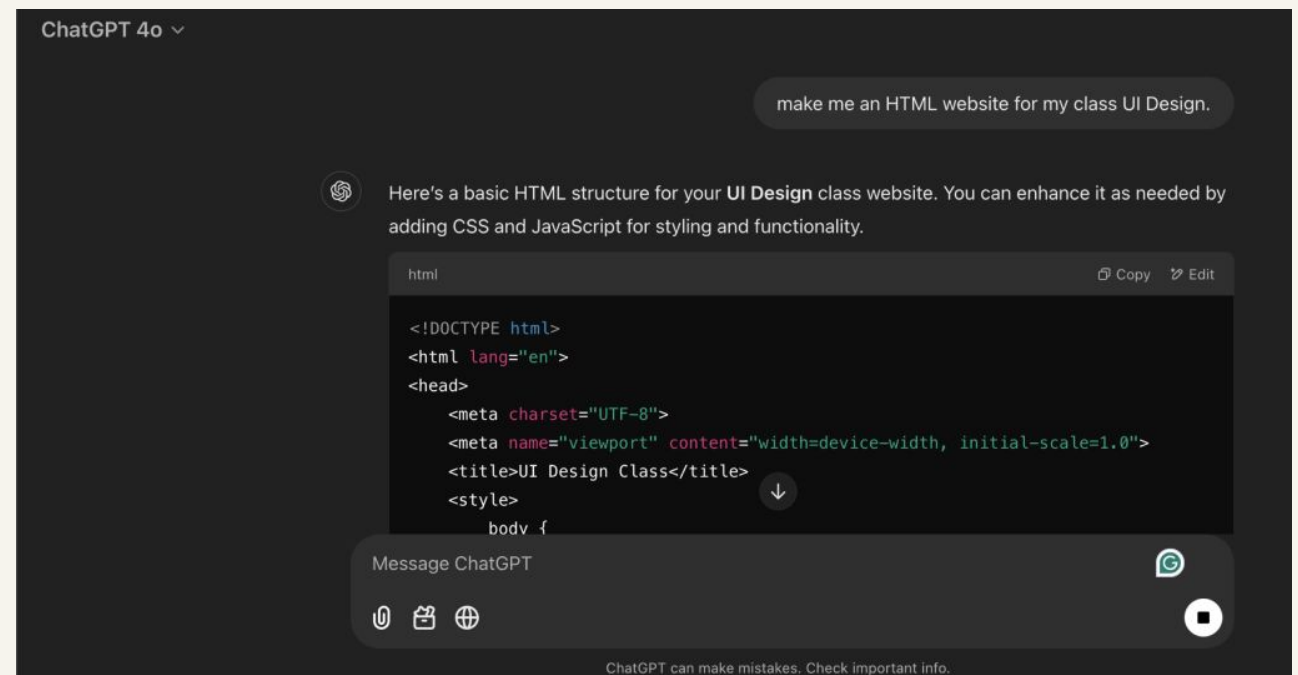
Even better than good error messages is a careful design which prevents a problem from occurring in the first place.

VIOLATION



6. Recognition rather than recall

Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another.



6. Recognition rather than recall

ChatGPT 4o ▾

make me an HTML website for my class UI Design.

 Here's a basic HTML structure for your **UI Design** class website. You can enhance it as needed by adding CSS and JavaScript for styling and functionality.

```
html
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>UI Design Class</title>
  <style>
    body {
```

Message ChatGPT 

ChatGPT can make mistakes. Check important info.





Top-rated cookware & kitchen tools



Hi, Alex
Customer since 2006

Your recent order



[See your orders](#)

Recently viewed



Today

[Edit your browsing history](#)

Related to items you've viewed



[Bundle and save](#)

Whole Foods Market

Free grocery delivery from your local store to your door

[Shop now](#)



JACK & JILL
More absorbent with leak secure protection

Jack & Jill Diapers, Size 4T, 162-count

★★★★★ 1,133

\$35.99 prime

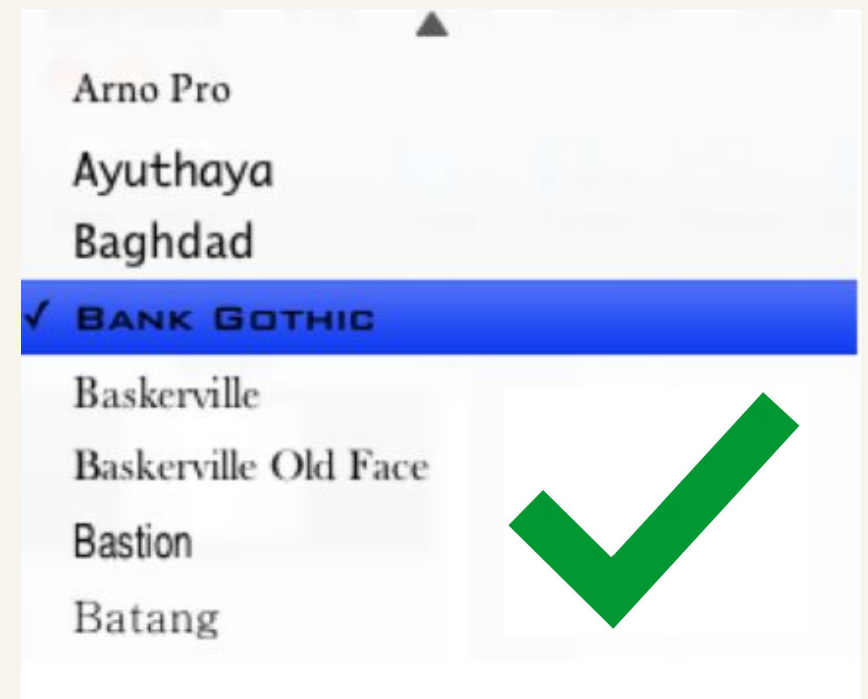
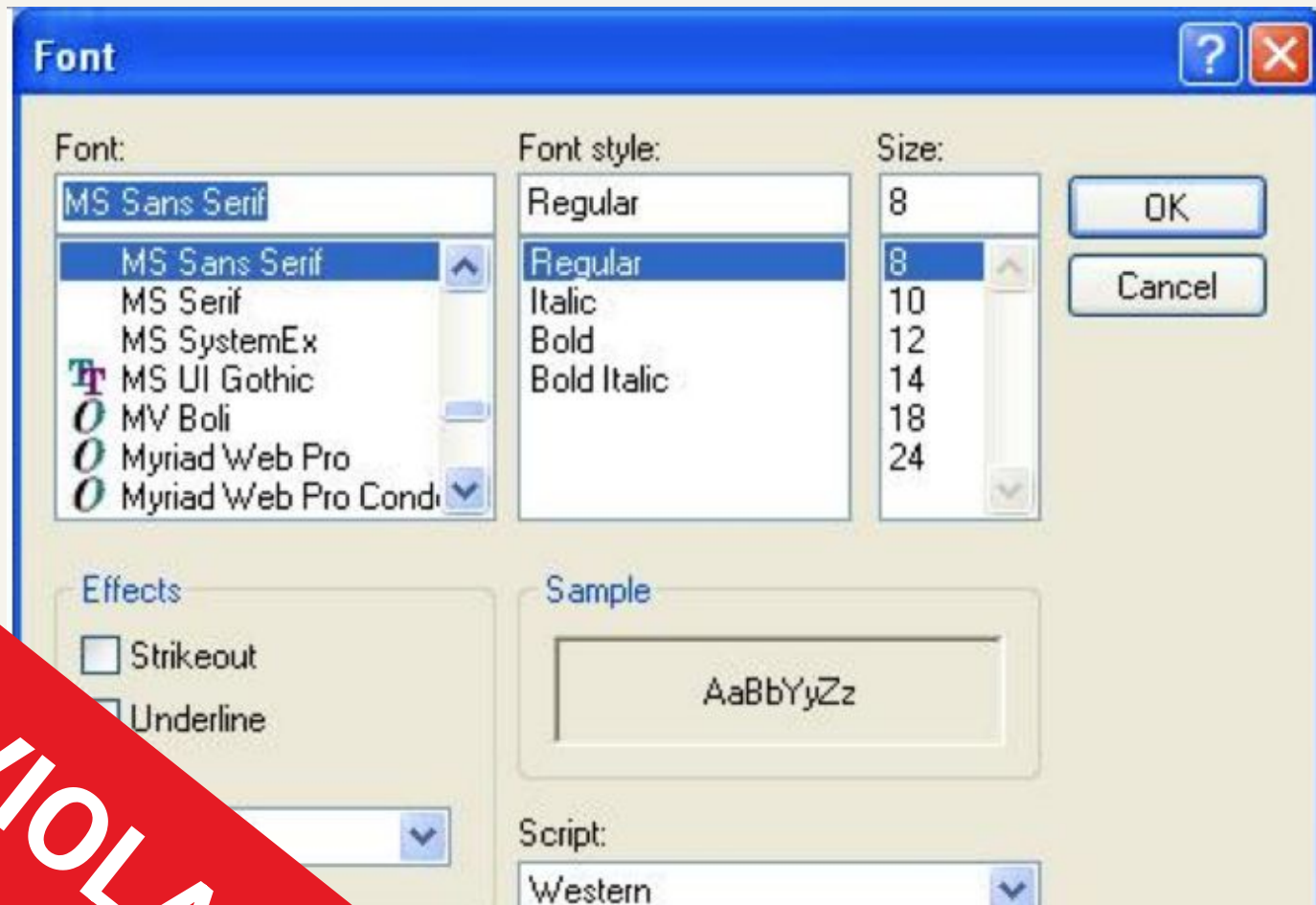
[Add to Cart](#)

Video: Recommended for you

Shop The Big Style Sale

Trade in a tablet device

6. Recognition rather than recall



VIOLATION

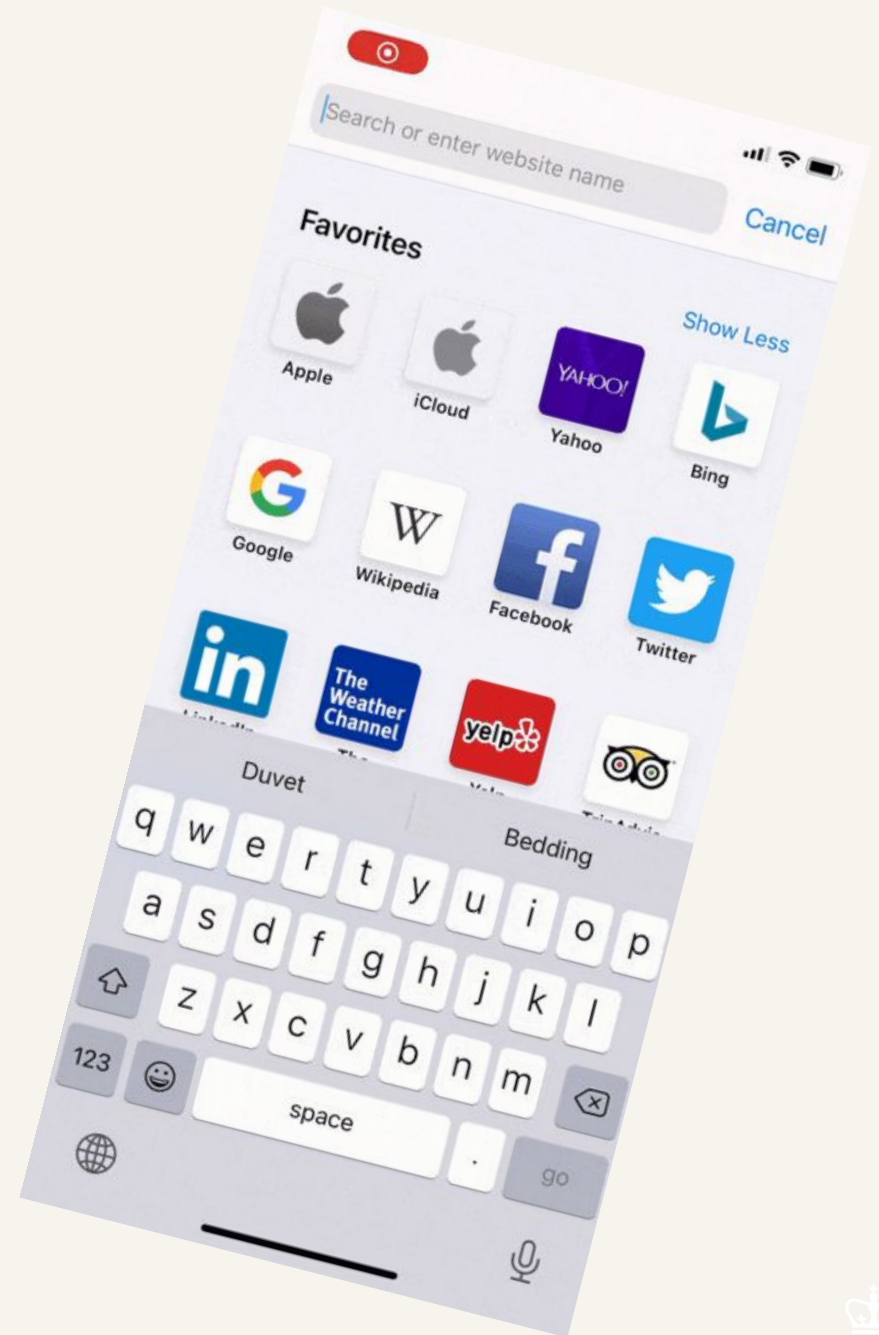
7. Flexibility and efficiency of use

Accelerators — unseen by the novice user — may often speed up the interaction for the expert. Allow users to tailor frequent actions.



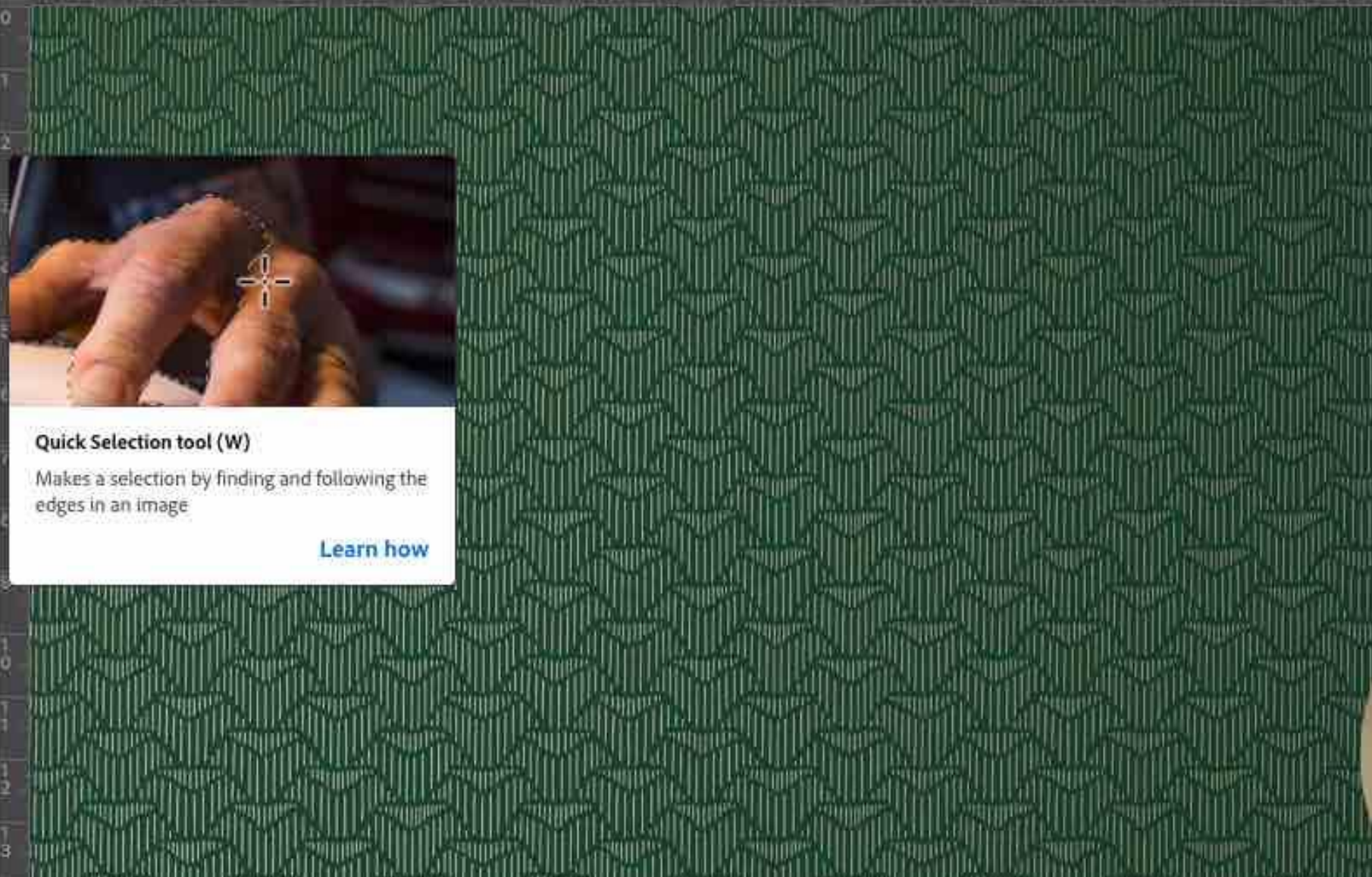
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HW_LF006_Palma_DeepGreen_Room_web.jpg @ 50% (RGB/8#)

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

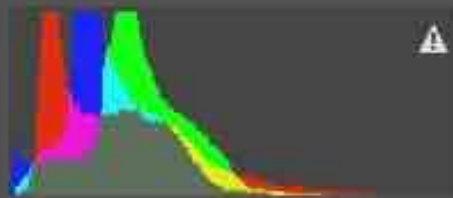


Quick Selection tool (W)

Makes a selection by finding and following the edges in an image

[Learn how](#)

Histogram Navigator



Libraries Adjustments

Search Adobe and Adobe Stock

My Library

View by type

Colors

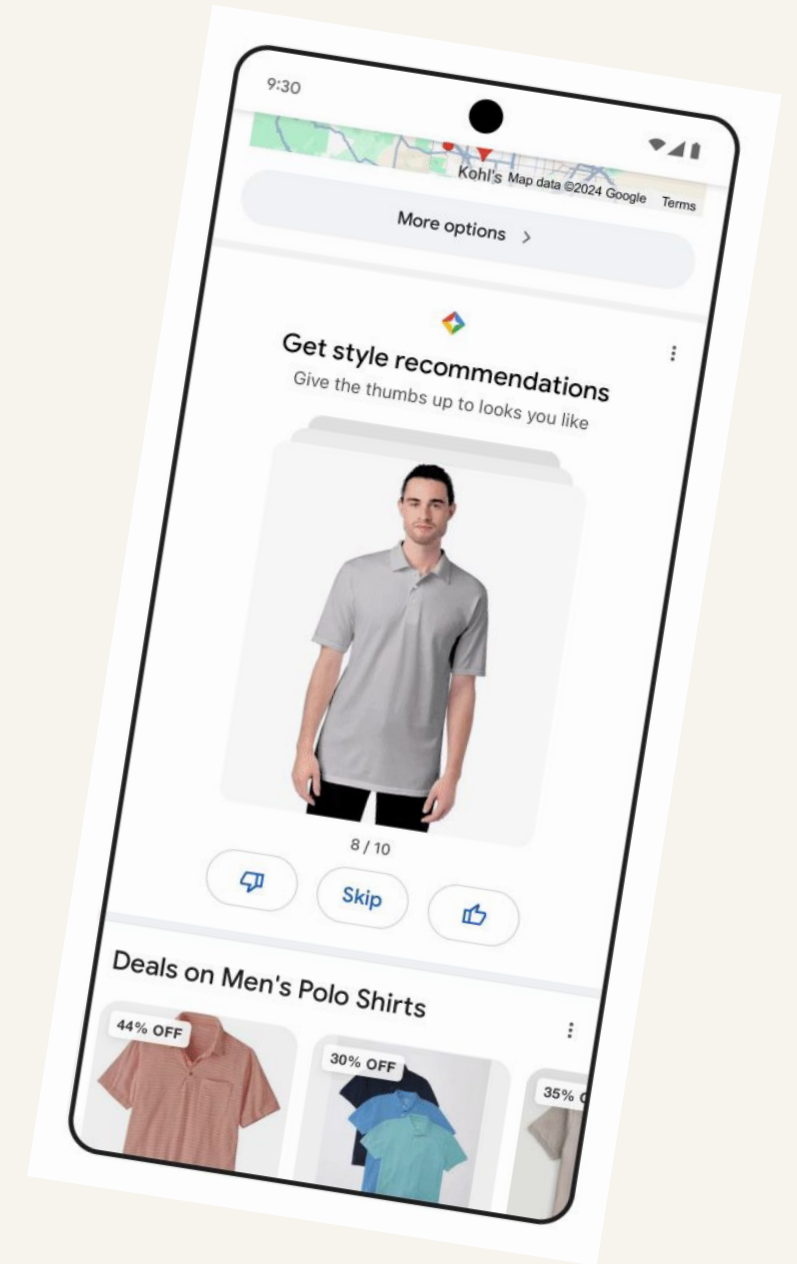


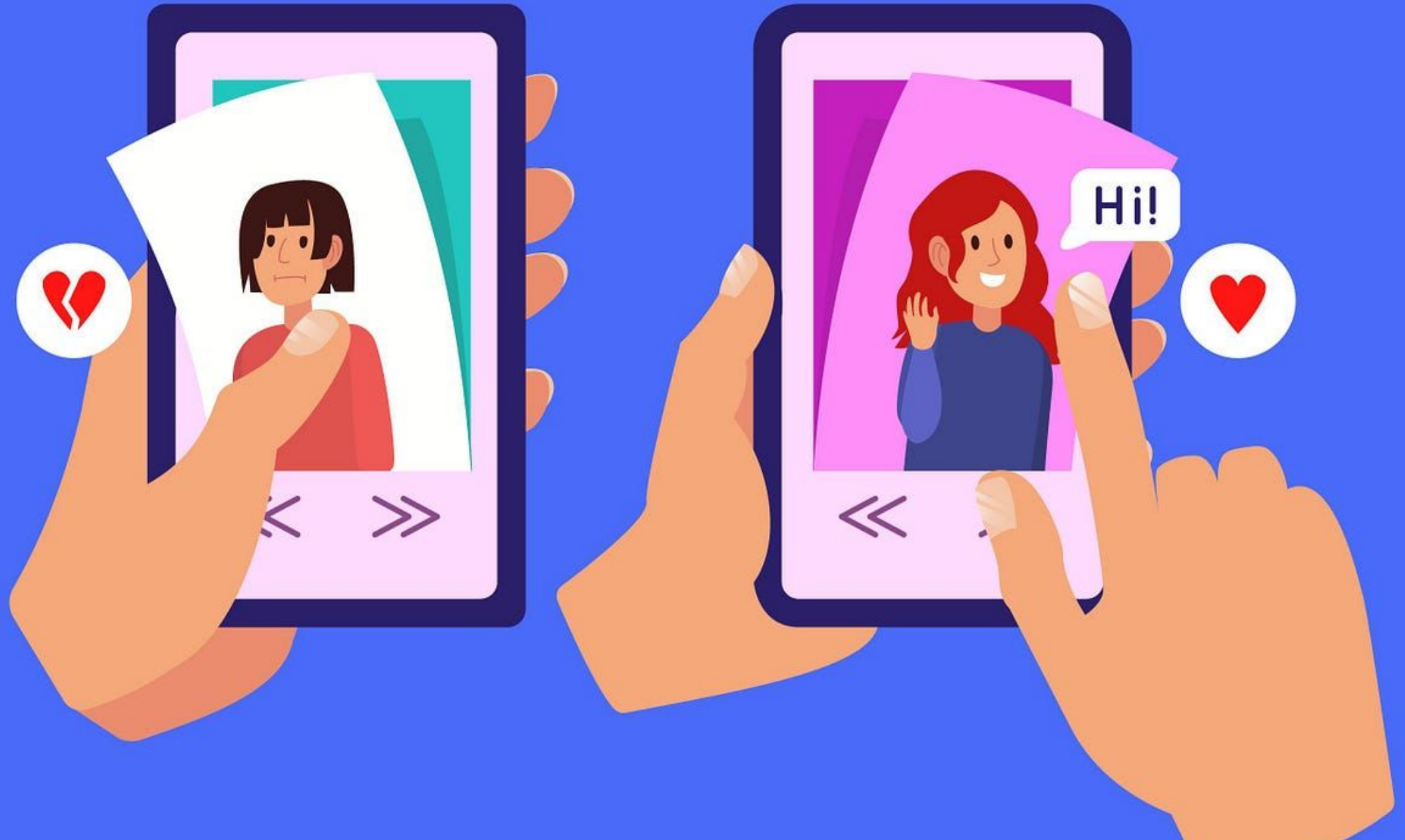
Color themes



8. Aesthetic and minimalist design

Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.







Google Search

I'm Feeling Lucky



Reservations Travel information Deals & offers MileagePlus® United Hub

Welcome to united.com

MileagePlus:
Sign in or join



Book travel



Flight

Hotel

Car

Cruise

Vacation

☐ Roundtrip

☐ One-way

[Multi-city](#)

From*

To*

☐ My dates are flexible

Depart date*



Return date*



1 adult



Economy



☐ Search for award travel

[Changed bag rules and optional services](#)

☐ Nonstop

[All search options](#)

Search

Flight status



Check-in



My trips



50,000 BONUS
MILES



United ClubSM
membership

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Settings

[Find your next travel destination](#)

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Products & services

Popular destinations

Important notices

Corporate information

Media & partners

Business services

VIOLATION



9. Help users recognize, diagnose, and recover from errors

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

Enter your contact information

Email Address

⚠ Please enter a valid email address.

Phone Number

⚠ Please enter a phone number.

☐ Send me text notifications regarding order updates. (Mobile phone only)

Continue to Payment Information

Choose a username (no spaces)

⚠ bert is already taken. Please choose a different username.

Choose a password

⚠ Passwords must be at least 6 characters and can only contain letters and numbers.

Retype password

Email address (must be real!)

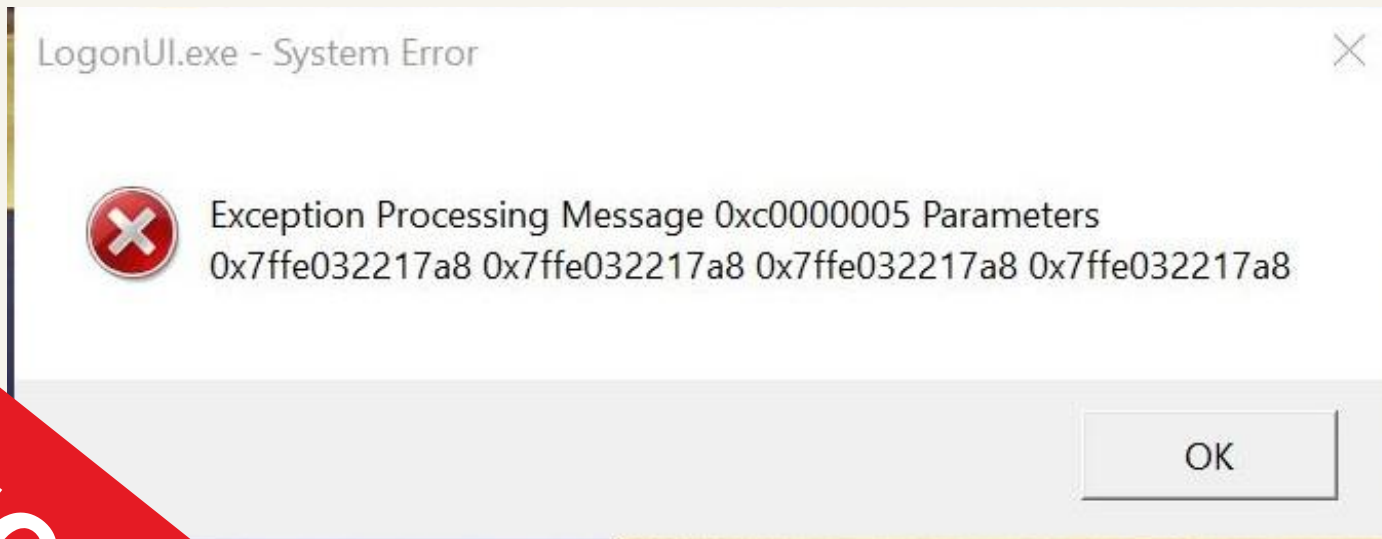
⚠ The email provided does not appear to be valid

☒ Send me occasional Digg updates.



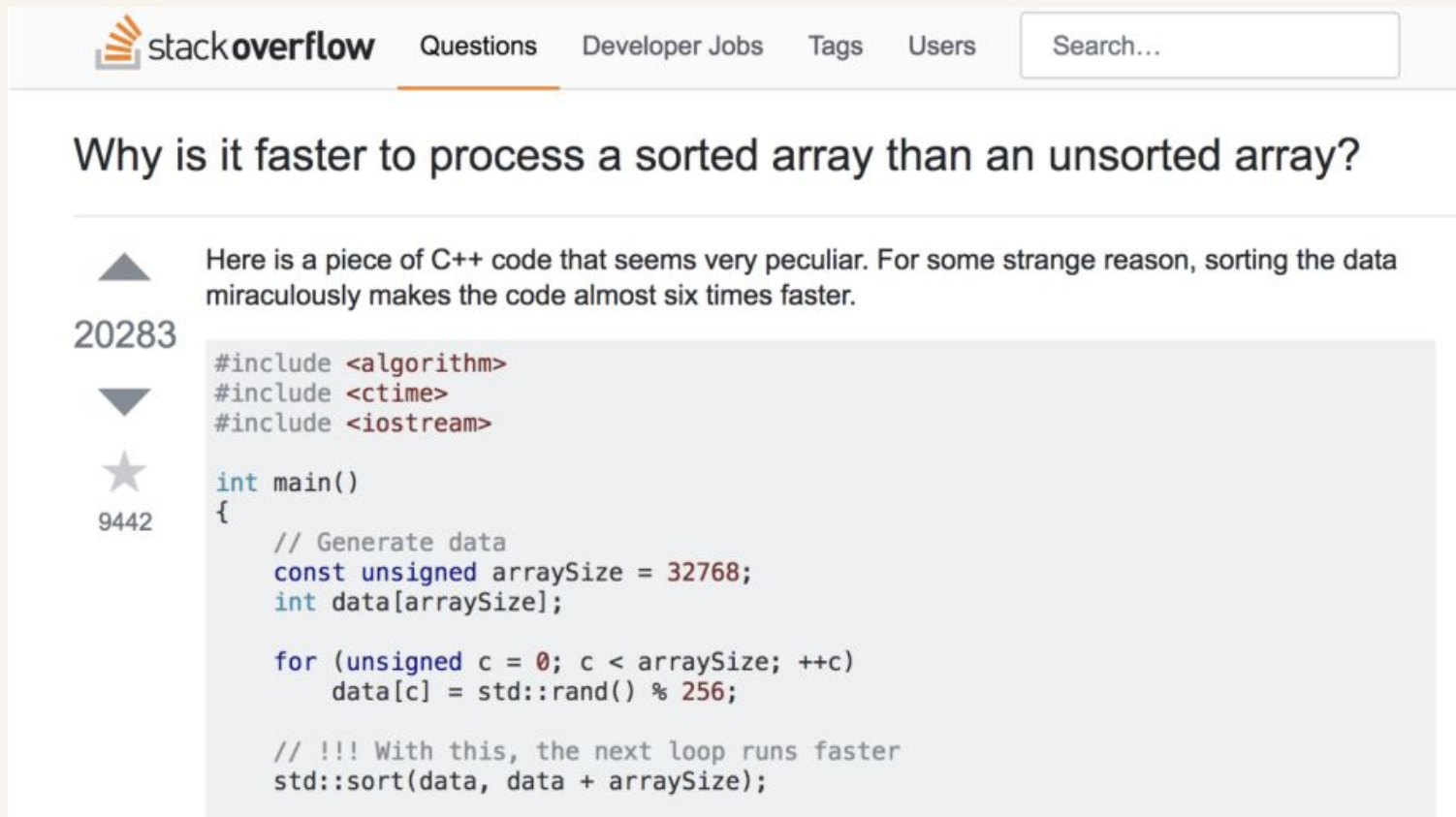
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10. Help and documentation

Documentation should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.



The screenshot shows the Stack Overflow website interface. At the top, there is a navigation bar with the Stack Overflow logo, links for 'Questions', 'Developer Jobs', 'Tags', and 'Users', and a search bar labeled 'Search...'. Below the navigation bar, the main heading of the question is 'Why is it faster to process a sorted array than an unsorted array?'. To the left of the question text, there are three icons: an upward-pointing triangle, a downward-pointing triangle, and a star. Next to these icons are the numbers '20283' and '9442' respectively. The question text reads: 'Here is a piece of C++ code that seems very peculiar. For some strange reason, sorting the data miraculously makes the code almost six times faster.' Below the text, there is a code block containing C++ code. The code includes headers for <algorithm>, <ctime>, and <iostream>. It defines a main function that generates an array of 32768 integers, each with a random value between 0 and 255. A comment indicates that the next loop runs faster after sorting the array.

```
#include <algorithm>
#include <ctime>
#include <iostream>

int main()
{
    // Generate data
    const unsigned arraySize = 32768;
    int data[arraySize];

    for (unsigned c = 0; c < arraySize; ++c)
        data[c] = std::rand() % 256;

    // !!! With this, the next loop runs faster
    std::sort(data, data + arraySize);
```





Windows 95 Video Guide with Matthew Perry & Jennifer Aniston

216,782 views

2K 48 SHARE

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Published on Dec 12, 2016

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Subscribe and watch scenes from all seasons of Friends on my channel - <https://www.youtube.com/channel/UCA8V...>

SHOW MORE

Up next

AUTOPLAY

Goes



Would I Lie to You?
(NEW WILTY 19th)
David Mitchell
Recommended for you
New



Actor Replacements That Totally Ruined The Show
Looper
3M views



Inside Matthew Perry's Tragic Real Life Story
Nicki Swift
952K views



Friends - ET On VH1: Behind The Friends Finale
Favorite Videos
1M views



TEENS REACT TO WINDOWS 95
FBE
12M views



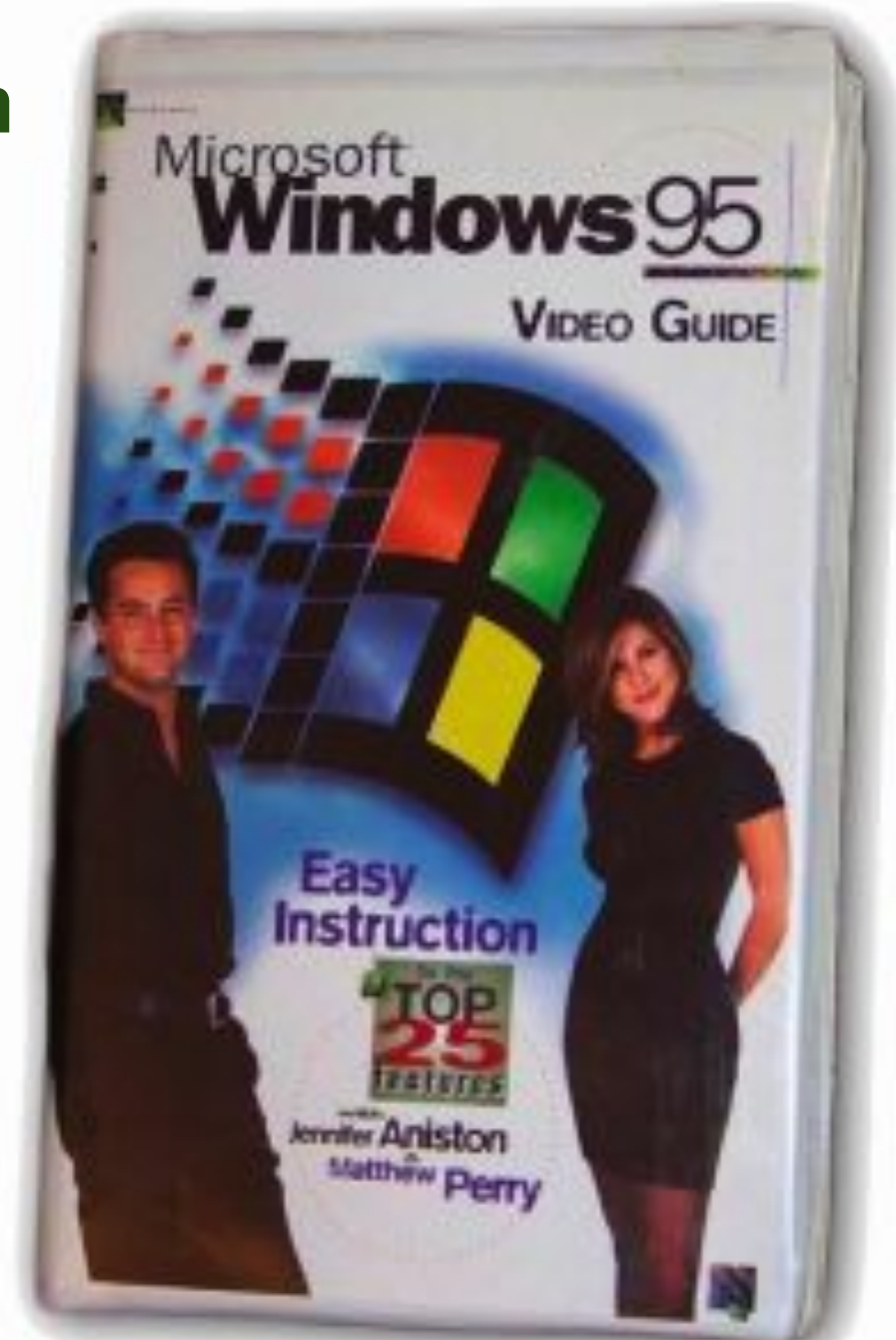
Friends - Ross and Monica's Cousin
Favorite Videos

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Nielsen's 10 Usability Heuristics

1. Visibility of system status
2. Match the real world
3. User control and freedom
4. Consistency and Standards
5. Error prevention
6. Recognition rather than recall
7. Flexibility and efficiency of use
8. Aesthetic and minimalist design
9. Recover from Errors
10. Help and documentation



Lecture 01: 10 Usability Heuristics

Homework 01



Homework 01

Warm Up due **Friday at 11:59PM**

- You can complete this now

Main Assignment due **Monday at 11:59PM**

- You will need Tuesday's lecture

Homework is posted to Courseworks:
<https://courseworks2.columbia.edu/courses/237745>



